

ML ASSIGNMENT-3

Q1)

given:

data points:

A(1,2), B(2,2), C(4,4), D(5,4), E(5,5)

initial centroids:

C1 = (1,2) C2 = (2,2)

clusters: k=2

distance Metric: Manhattan Distance

Step 1: Assigning each point to the nearest centroid

$$\text{distance} = |x_1 - x_2| + |y_1 - y_2|$$

using the above formula

given datapoint	distance to C1(1,2)	distance to C2=(2,2)	assigned cluster
A(1,2)	0	$ 1-2 + 2-2 = 1$	C1
B(2,2)	$ 1-2 + 2-2 = 1$	0	C2
C(4,4)	$ 4-1 + 4-2 = 5$	$ 4-2 + 4-2 = 2$	C2
D(5,4)	$ 5-1 + 4-2 = 6$	$ 5-2 + 4-2 = 5$	C2
E(5,5)	$ 5-1 + 5-2 = 7$	$ 5-2 + 5-2 = 6$	C2

Hence,

Cluster assignments:

Cluster 1: A(1,2)

Cluster 2: B(2,2), C(4,4), D(5,4), E(5,5)

Step 2: Updating Centroids

In Manhattan we use median for center:

Cluster 1: Only point is A(1,2) $\rightarrow$ centroid remains same

Cluster 2:

$$X_s: 2, 4, 5, 5 \rightarrow \text{median} = (4+5)/2 = 4.5$$

$$Y_s: 2, 4, 4, 5 \rightarrow \text{median} = (4+4)/2 = 4$$

$$\text{new centroid} = (4.5, 4)$$

Step 3: Reassigning points using new centroids

recalculating distances to updated centroids:

datapoint	distance to C_1	distance to C_2	Assigned cluster
A(1,2)	0	$ 1-4.5 + 1-4 = 6.5$	C_1
B(2,2)	$ 2-1 + 2-2 = 1$	$ 2-4.5 + 2-4 = 3.5$	C_1
C(4,4)	$ 4-1 + 4-2 = 5$	$ 4-4.5 + 4-4 = 0.5$	C_2
D(5,4)	$ 5-1 + 4-2 = 6$	$ 5-4.5 + 4-4 = 0.5$	C_2

$$E(5,5)$$

$$|5-1| + |5-2|$$

$$= 7$$

$$|5-4.5| + |5-4|$$

$$= 2.5$$

C2

Hence

cluster assignments:

Cluster 1: A(1,2), B(2,2)

Cluster 2: ~~A(1,2)~~, C(4,4), D(5,4), E(5,5)

Repeating Step 2: Updating centroids

Cluster 1:

X_s : 1, 2 $\rightarrow$ median = 1.5

Y_s : 2, 2 $\rightarrow$ median = 2

new centroid: (1.5, 2)

Cluster 2:

X_s : 4, 5, 5 $\rightarrow$ median = 5

Y_s : 4, 4, 5 $\rightarrow$ median = 4

new centroid: (5, 4)

Repeating Step 3: $|p-2| + |2 \cdot p - 2|$

data points

cluster distance to C_1

distance to C_2

cluster assignments

A(1,2)

$$|1-1.5| + |2-2| = 0.5$$

$$|1-5| + |2-4| = 6$$

C_1

B(2,2)

$$|2-1.5| + |2-2| = 0.5$$

$$|2-5| + |2-4| = 5$$

C_1

C(4,4)

$$|4-1.5| + |4-2| = 2.5 + 2 = 4.5$$

$$|4-5| + |4-4| = 1$$

C_2

D(5,4)

$$|5-1.5| + |4-2| = 3.5 + 2 = 5.5$$

$$|5-5| + |4-4| = 0$$

C_2

E(5,5)

$$|5-1.5| + |5-2| = 3.5 + 3 = 6.5$$

$$|5-5| + |5-4| = 1$$

C_2

Hence new cluster assignment:

$$C_1 = A(1,2), B(2,2)$$

$$C_2 = C(4,4), D(5,4), E(5,5)$$

The cluster assignment does not change

They show convergence and the algorithm

stops

Q2:

Step 1: Entropy of the given dataset

We have

high: 6

low: 2

$$\text{Entropy}(S) = -\frac{6}{8} \log_2\left(\frac{6}{8}\right) - \frac{2}{8} \log_2\left(\frac{2}{8}\right)$$

$$= -0.75 \log_2(0.75) - 0.25 \log_2(0.25)$$

$$= 0.811$$

Step 2: Calculating information gain for each attribute

for attribute age:

$<25 \rightarrow 5$ examples: All high

$>25 \rightarrow 3$ examples: 1 high, 2 low

entropy(<25) = 0 (pure)

$$\text{Entropy}(Z25) = -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \approx 0.918$$

$$\text{gain}(\text{age}) = 0.811 - \left(\frac{5}{8} \cdot 0 + \frac{3}{8} \cdot 0.918 \right)$$

$$= 0.811 - 0.344$$

$$= 0.467$$

for attribute income

$>50K \rightarrow$ 4 examples: 2 high, 2 low

$<50K \rightarrow$ 4 examples: all high

$$\text{Entropy } <50K = 0$$

$$\text{Entropy } >50K = -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right)$$

$$= -0.5 \cdot 1 - 0.5 \cdot 1$$

$$= 1.0$$

$$\text{gain}(\text{income}) = \text{Entropy}(S) - \text{Entropy}_{\text{split}} = 0.811 - 0.5$$

$$= 0.311$$

$$\text{Entropy}_{\text{split}} = \frac{4}{8} \cdot 1 + \frac{4}{8} \cdot 0 = 0.5$$

for attribute gender

M $\rightarrow$ 4 examples: 3 high, 1 low

F $\rightarrow$ 4 example: 3 high, 1 low

$$\text{Entropy}(M) = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \approx 0.811$$

$$\text{Entropy}(F) = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \approx 0.811$$

$$\text{Gain}(\text{gender}) = 0.811 - \left(\frac{4}{8} \cdot 0.811 + \frac{4}{8} \cdot 0.811 \right)$$

$$= 0.811 - 0.811$$

$$= 0$$

Step 3: Choosing best attribute to split

$$\text{age} = 0.467$$

$$\text{income} = 0.311$$

$$\text{gender} = 0$$

Hence we choose age to split

Step 4: building subtrees

Age < 25

for left node all samples are same no need to split

for right node (age > 25) ~~3 samples have high~~

we have 3 samples

1 high and 2 low

Entropy ≈ 0.918

so we do sub-gain for income and gender

we split using income as it has best gain

as ~~ear~~ proved below

split by income

$< 50k$ entropy $= 0$

$> 50k$ entropy $= -\log_2 1 = 0$

sum of entropy $= 0$

information gain $= 0.91$

split by gender

male entropy $= -1\log_2 1 = 0$

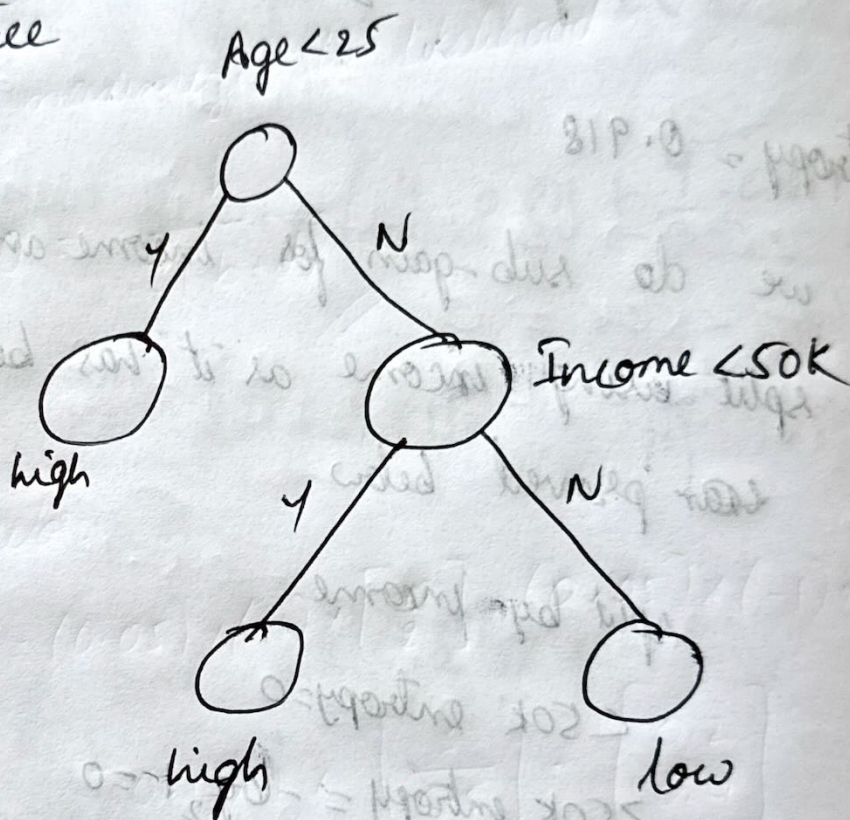
entropy(F) $= -0.5\log_2 0.5 - 0.5\log_2 0.5 = 1$

sum of entropies proportionally $0.33 \times 0 + 0.67 \times 1 = 0.67$

gain (gender) = 0.24

Hence the second decision node for eight child node is income

final decision tree



Q47 1. Suppose the number of A, C, G, or T is a, b, c, d respectively.

Likelihood: $p(a, b, c, d | \theta) = c \left(\frac{1}{3}\right)^a \theta^b \left(\frac{1}{3}\right)^c \left(\frac{1}{3} - \theta\right)^d$

$$\ln p(a, b, c, d | \theta) = \ln C + a \ln \frac{1}{3} + b \ln \theta + c \ln \frac{1}{3} + d \ln \left(\frac{1}{3} - \theta \right)$$

Maximize the likelihood (derivative = 0)

$$\frac{b}{\theta} - \frac{d}{\frac{1}{3} - \theta} = 0 \Rightarrow \theta = \frac{b}{3(b+d)}$$

2. Expectation: if we know the value θ , we can estimate the expected value of b and d :

$$\frac{a}{b} = \frac{1}{30}$$

$$\Rightarrow \frac{a+b}{b} = \frac{1+30}{30} (1) \times (20.0) = \sum_{i=1}^n \omega_i = \omega$$

$$\Rightarrow \frac{x}{b} = \frac{1+30}{30}$$

$$\Rightarrow b = \frac{30}{1+30} x$$

$$I = a \left(x - b \right) \times [n.o, s.o] + (1) \Leftarrow$$

$$\frac{c}{d} = \frac{1}{1-3\theta} \Rightarrow \frac{c+d}{d} = \frac{2-3\theta}{1-3\theta} \Rightarrow \frac{y}{d} = \frac{2-3\theta}{1-3\theta}$$

$$\Rightarrow d = \frac{1-3\theta}{2-3\theta} y$$

$$c = y - d$$

Maximization: when we know b and d , then we can find θ maximize the likelihood

$$\theta = \frac{b}{3(b+d)}$$

Q5) 1. Weight vector:

$$w = \sum_{i=1}^N \alpha_i y_i x_i = (0.05) \times (-1) \times \begin{bmatrix} 0 \\ 2 \end{bmatrix} + (0.05) \times (-1) \times \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 0.1 \times 1 \times \begin{bmatrix} 3 \\ 5.5 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

2. Select one support vector $x_1 = [0, 2]^T$

$$y_n (w^T x_n + b) = 1$$

$$\Rightarrow (-1) \times ([0.2, 0.4] \times \begin{bmatrix} 0 \\ 2 \end{bmatrix} + b) = 1$$

$$\Rightarrow b = -1.8$$

3. For the 3 closest points, the value $|w^T x + b|$ are normalized, and equals to 1, so the margin will be calculated by:

$$\text{Margin} = \frac{|w^T x + b|}{\|w\|} = \frac{1}{\|w\|} = \frac{1}{\sqrt{0.2}} = 2.236$$